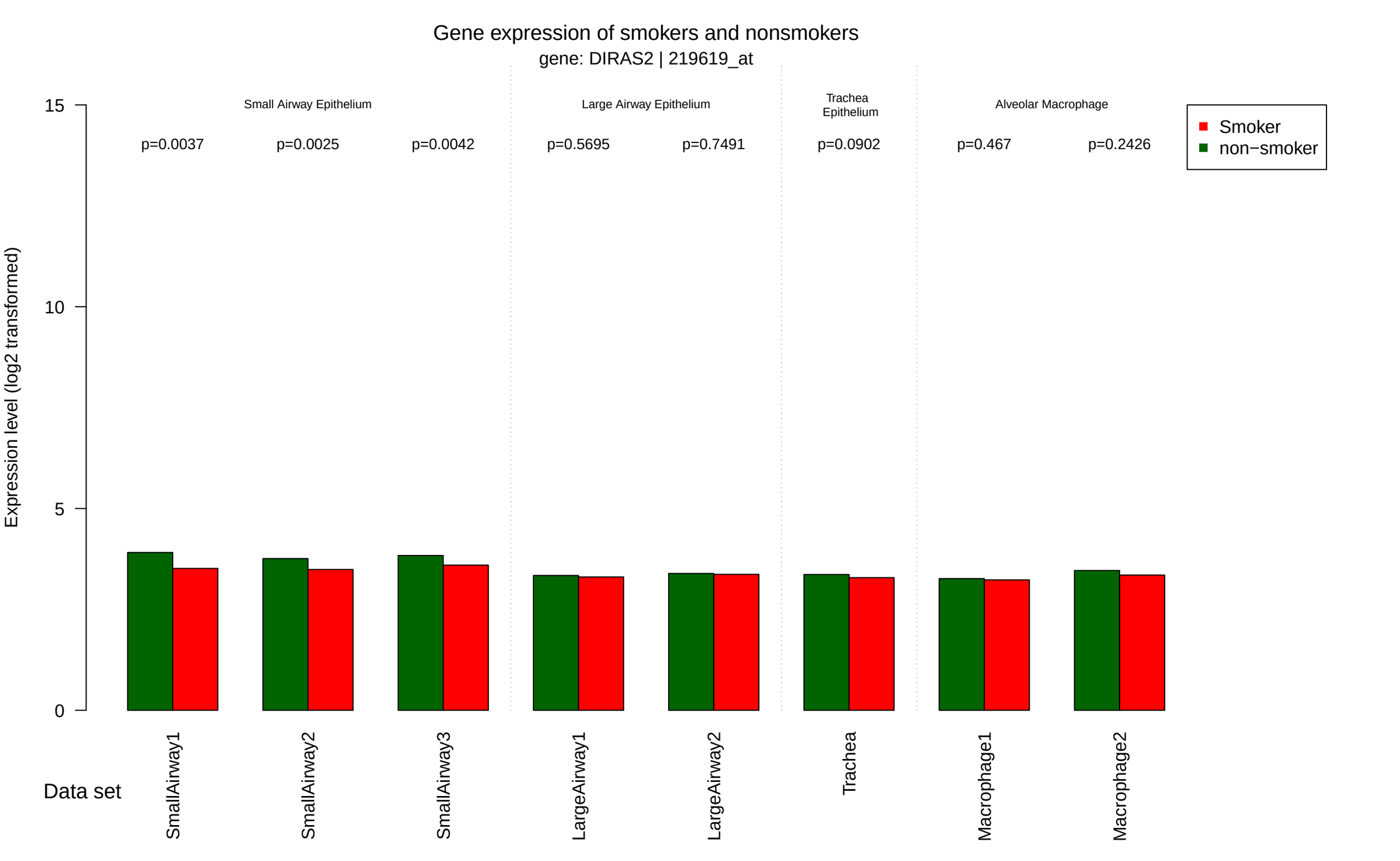
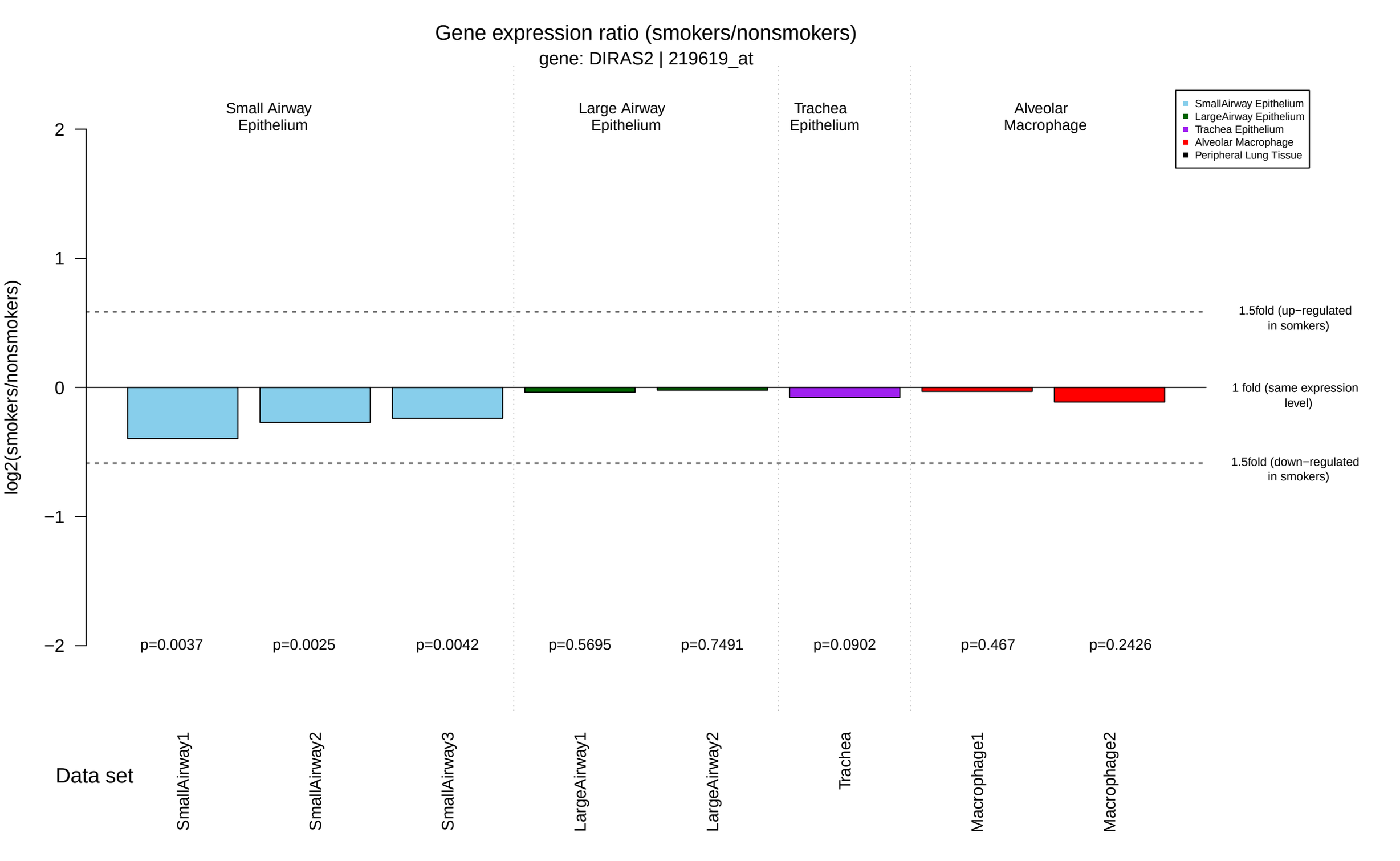
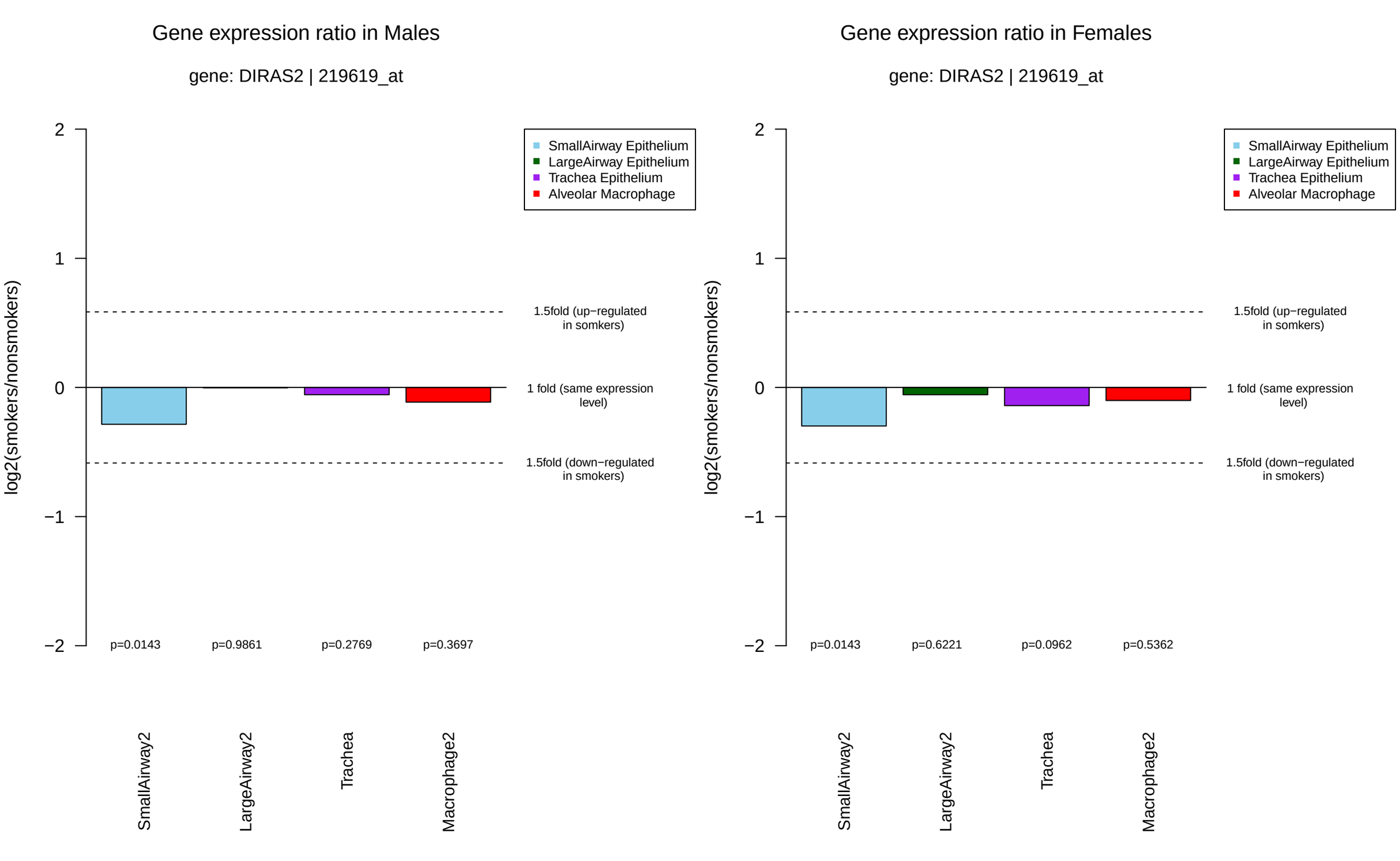
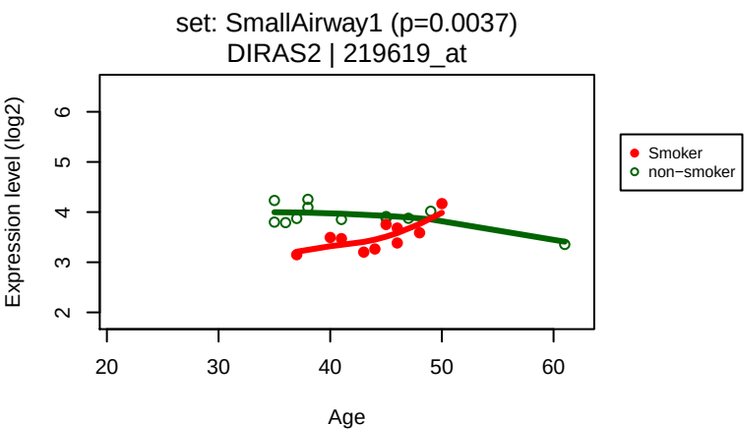
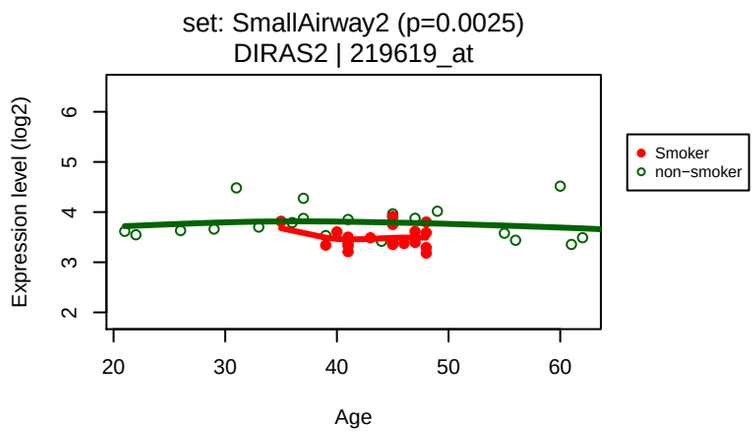
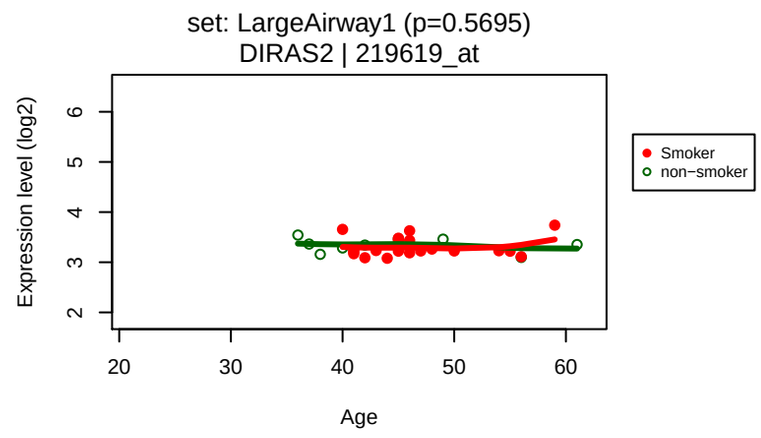

Gene expression of smokers and nonsmokers
gene: DIRAS2 | 219619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0037
p=0.0025
p=0.0042
p=0.5695
p=0.7491
p=0.0902
p=0.467
p=0.2426
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

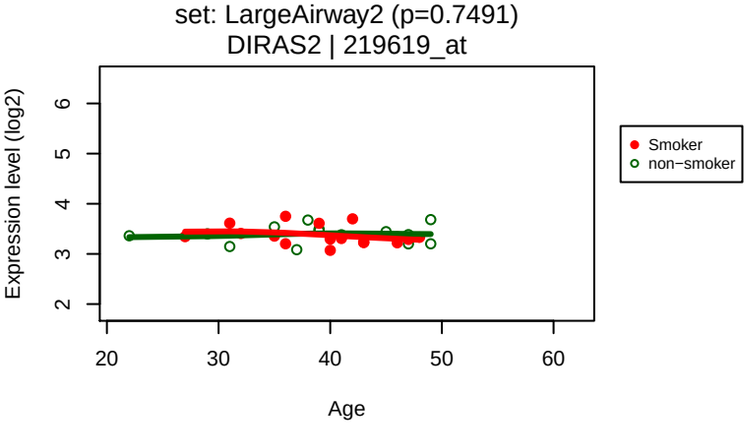

Gene expression ratio (smokers/nonsmokers)
gene: DIRAS2 | 219619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.0025
p=0.0042
p=0.5695
p=0.7491
p=0.0902
p=0.467
p=0.2426
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

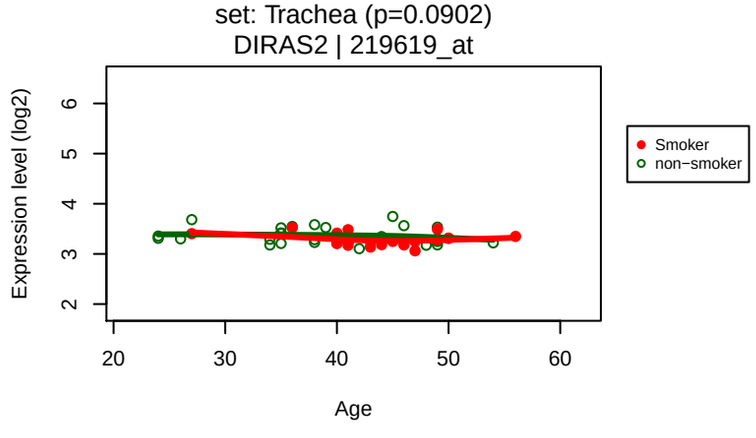

Gene expression ratio in Males
gene: DIRAS2 | 219619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0143
p=0.9861
p=0.2769
p=0.3697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DIRAS2 | 219619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0143
p=0.6221
p=0.0962
p=0.5362
SmallAirway2
LargeAirway2
Trachea
Macrophage2

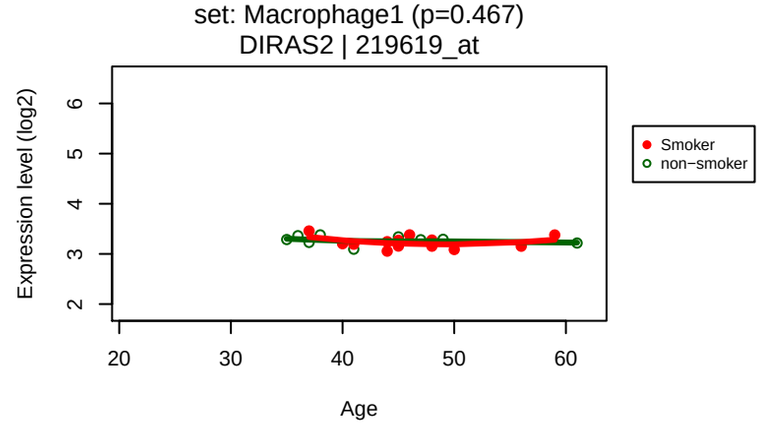

set: SmallAirway1 (p=0.0037)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker

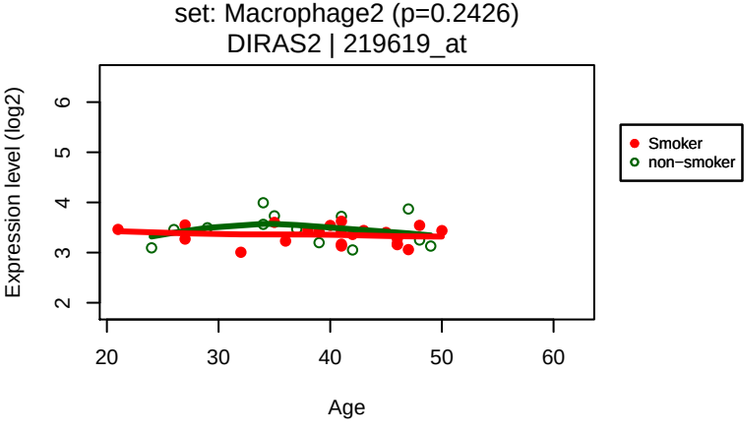

set: SmallAirway2 (p=0.0025)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker

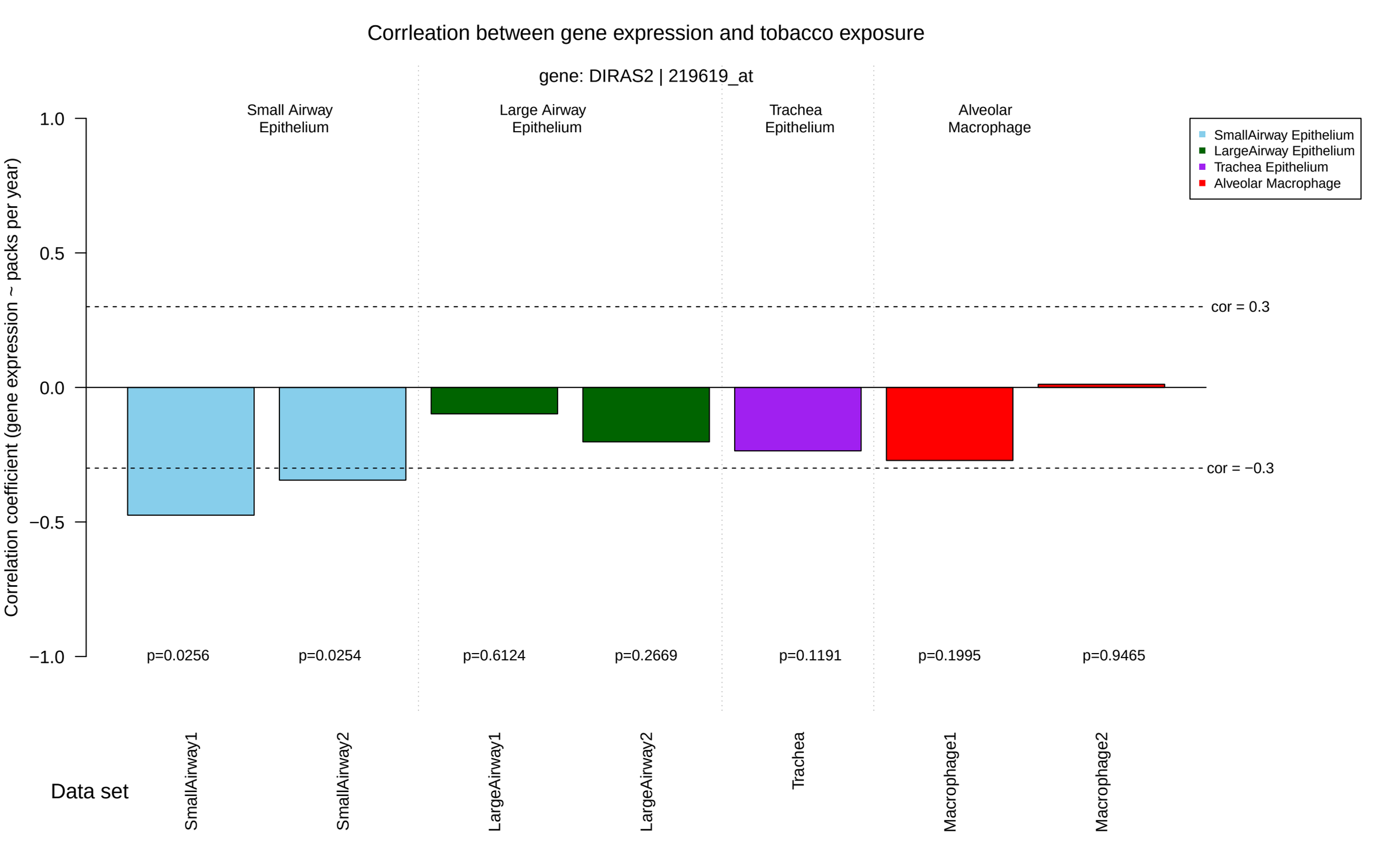

set: LargeAirway1 (p=0.5695)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7491)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0902)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.467)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2426)
DIRAS2 | 219619_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DIRAS2 | 219619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0256
p=0.0254
p=0.6124
p=0.2669
p=0.1191
p=0.1995
p=0.9465
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2